Azure Data Factory:

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1. What is Azure Data Factory?

Azure Data Factory (ADF) is a cloud-based data integration service that allows you to create, schedule, and orchestrate workflows for data movement and transformation. It supports ETL (Extract, Transform, Load) and ELT (Extract, Load, Transform) processes across various data sources.

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2. What are the key components of Azure Data Factory?

- **Pipelines:** Logical grouping of activities.
- **Activities:** Tasks within a pipeline (e.g., Copy, Transform).
- **Linked Services:** Connection strings to data sources/destinations.
- **Datasets:** Structures representing data used in activities.
- **Triggers:** Schedule or event-based pipeline executions.
- **Integration Runtime (IR):** Computes environment for activities.

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3. What is a pipeline in Azure Data Factory?

A pipeline is a logical grouping of activities that perform tasks such as copying data, transforming it, or controlling workflows.

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4. What is an activity in Azure Data Factory?

An activity is a task within a pipeline, like copying data, executing a stored procedure, or running a transformation in Databricks.

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5. What are linked services in Azure Data Factory?

Linked services define connections to data stores or compute resources. For example, a linked service can connect to an Azure Blob Storage or an SQL Database.

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6. What are datasets in Azure Data Factory?

Datasets define the structure of the data within linked services. For instance, a dataset might define a CSV file stored in Blob Storage.

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7. What are triggers in Azure Data Factory?

Triggers are mechanisms to execute pipelines. Types include schedule-based, tumbling window, and event-based triggers.

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8. Explain the difference between a pipeline and a dataset.

- **Pipeline:** Orchestrates activities.
- **Dataset:** Represents the structure of data processed by those activities.

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9. What are the types of triggers available in Azure Data Factory?

- **Schedule triggers:** Run pipelines at a scheduled time.
- **Tumbling window triggers:** Process data in periodic intervals.
- **Event-based triggers:** React to blob events (e.g., file arrival).

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10. What is the integration runtime (IR) in Azure Data Factory?

IR is the compute infrastructure used by ADF for data movement, data transformation, and activity execution.

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11. What are the types of integration runtime in Azure Data Factory?

1. **Azure IR:** Cloud-based for data movement and transformation.
2. **Self-hosted IR:** On-premises or hybrid scenarios.
3. **Azure-SSIS IR:** For running SSIS packages in ADF.

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12. What is the Copy Data activity?

The Copy Data activity moves data from a source to a destination, supporting data transformation during the process.

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13. What is the purpose of the ForEach activity?

ForEach activity loops through an array of values, executing inner activities for each item.

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14. What is the use of the Filter activity in Azure Data Factory?

The Filter activity filters a collection based on conditions, passing only the matching items to subsequent activities.

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15. What is Azure Data Factory used for in real-world scenarios?

- Migrating data to cloud storage.
- Building ETL/ELT pipelines.
- Ingesting IoT data in real-time.
- Orchestrating data processing workflows.

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16. How does ADF ensure data transformation?

ADF uses Mapping Data Flows for no-code transformations or external compute like Databricks and Azure Functions for custom logic.

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17. Can you explain the concept of control flow in Azure Data Factory?

Control flow orchestrates pipeline execution using activities like Execute Pipeline, If Condition, and Wait.

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18. What are self-hosted integration runtimes?

A self-hosted IR is installed on-premises to enable secure data movement and activity execution in private networks.

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19. What are tumbling window triggers?

Tumbling window triggers process data at fixed time intervals, ensuring no overlap and maintaining data consistency.

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20. What is the difference between event-based triggers and schedule-based triggers?

- **Event-based triggers:** React to events like file arrival.
- **Schedule-based triggers:** Execute pipelines at specific times.

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21. How can you monitor pipeline execution in Azure Data Factory?

Use the Monitor tab in the ADF portal for real-time execution logs, set alerts, and analyze activity details.

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22. What are the default monitoring features available in ADF?

- Run history of pipelines.
- Activity-level logs.

- Alerts and diagnostic logging integration with Azure Monitor.

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23. What is the difference between Azure Data Factory and SSIS?

ADF is a cloud-native, scalable integration tool, whereas SSIS is an on-premises ETL tool that focuses on SQL Server ecosystems.

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24. Explain the purpose of the Lookup activity.

The Lookup activity retrieves data from a data source, often used for configuration or control flow.

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25. What is the role of parameters in Azure Data Factory?

Parameters allow dynamic values to be passed into pipelines, datasets, or linked services.

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26. How does Azure Data Factory handle error handling in pipelines?

ADF offers retry policies, failure path logic in activities, and logging for troubleshooting.

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27. What is the purpose of the Debug mode in Azure Data Factory?

Debug mode tests pipelines without publishing them, saving development time.

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28. What is the difference between dynamic content and parameters?

- **Dynamic content:** Expressions that compute values.
- **Parameters:** Static values passed to a pipeline.

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29. What are the types of authentication supported in Azure Data Factory?

- Azure AD-based authentication.
- Managed Identity.
- Service Principal.

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30. What is the purpose of the Get Metadata activity?

The Get Metadata activity retrieves metadata (e.g., file size, last modified date) from data sources.

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31. What is data integration in Azure Data Factory?

Data integration combines data from multiple sources for processing, transformation, or analysis.

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32. Can Azure Data Factory handle unstructured data? If yes, how?

Yes, by integrating with storage like Blob Storage or Data Lake and using activities like Copy or Data Flows.

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33. What are the best practices for designing pipelines in Azure Data Factory?

- Modular design using reusable pipelines.
- Secure connections with Managed Identity.
- Enable logging and monitoring.

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34. How can you implement incremental data loading in Azure Data Factory?

By filtering new or changed records using watermark columns or Change Data Capture (CDC).

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35. What are sinks and sources in Azure Data Factory?

- **Source:** The origin of data.
- **Sink:** The destination of data.

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36. Explain the difference between Azure Data Lake and Azure Data Factory.

Data Lake stores data; Data Factory moves and transforms it.

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37. How does Azure Data Factory integrate with other Azure services?

ADF integrates with Blob Storage, Synapse Analytics, Azure Functions, Databricks, and more.

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38. What are the limitations of Azure Data Factory?

- Limited support for complex transformations compared to Databricks.
- Debugging large pipelines can be complex.

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39. How can you secure data in Azure Data Factory?

- Use Managed Identity and RBAC.

- Encrypt data in transit and at rest.
- Leverage private endpoints.

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40. Explain the role of Azure Key Vault in Azure Data Factory.

Key Vault stores secrets, keys, and credentials securely, which ADF can access via linked services.

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41. What is the concept of parallelism in Azure Data Factory?

Parallelism enables concurrent execution of activities to improve pipeline performance.

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42. How does Azure Data Factory handle schema drift?

Using Mapping Data Flows or dynamic column mapping in Copy Activity to adjust for schema changes.

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43. What are the different ways to scale Azure Data Factory pipelines?

- Increase activity parallelism.
- Optimize integration runtime configurations.
- Use partitioned data processing.

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44. Explain the integration of Azure Data Factory with Databricks.

ADF pipelines can call Databricks notebooks for advanced transformations using Databricks activities.

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45. What is the role of Azure Monitor in Azure Data Factory?

Azure Monitor tracks pipeline performance, logs metrics, and generates alerts for failures.

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46. How can you automate Azure Data Factory pipeline deployment?

By using Azure DevOps CI/CD pipelines or ARM templates.

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47. What are the different logging mechanisms available in Azure Data Factory?

- Diagnostic logs.
- Activity-run logs.

- Integration with Log Analytics and Azure Monitor.

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